16. n = 6

According Aufbau principle,

6s →4f →5d →6p

ns → (n-2)f →(n-1)d → np

17 (ii) l can have the values from 0 to n-1

n=2; possible 'l' values are 0, 1 hence l = 2 is not possible.

(iv) for l=0; m=-1 not possible

(v) for n=3 l = 4 and m = 3 not possible

18. n+l = 8

Electronic configuration of atom with atomic number 105 is [Rn] $5f^{14}$ $6d^3$ $7s^2$

Orbital	(n+l)	No. of electrons
5f	5+3 = 8	14
6d	6+2 = 8	3
7s	7+0 = 7	2
No of electrons = 14 + 3 = 17		

19. Option (a) - Zero **(Refer to Figure 2.9)**

20.

$$\Delta x.\Delta p \geq \frac{h}{4\pi}$$

$$\Delta p.\Delta p \geq \frac{h}{4\pi}$$

$$\Delta p^2 \geq \frac{h}{4\pi}$$

$$m^2(\Delta v)^2 \geq \frac{h}{4\pi}$$

$$(\Delta v) \geq \sqrt{\frac{h}{4\pi m^2}}$$

$$\Delta v \geq \frac{1}{2m}\sqrt{\frac{h}{\pi}}$$

21.

m = 100 g = 100 x 10^{-3} kg

v = 100 cm s^{-1} = 100 x 10^{-2} m s^{-1}

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.626\times10^{34}\,Js^{-1}}{100\times10^{-3}\,kg\times100\times10^{-2}\,ms^{-1}}$$

$$= 6.626\times10^{-33}\,ms^{-1}$$

$$= 6.626\times10^{-31}\,cms^{-1}$$

23.

$$E_n = \frac{-13.6}{n^2}\,eV\ atom^{-1}$$

$$E_1 = \frac{-13.6}{(1)^2} = -13.6$$

$$E_3 = \frac{-13.6}{(3)^2} = -\frac{13.6}{9}$$

Givethat

$$E_3 = -E$$

$$-\frac{13.6}{9} = -E$$

$$-13.6 = -9E$$

$$\Rightarrow E_1 = -9E$$

II Key to brief answer question:

27. n = 4 l = 0, 1, 2, 3

four sub-shells ⇒ s, p, d, f

l = 0 m_l = 0; one 4s orbital.

l = 1 m_l = −1, 0, +1 ; three 4p orbitals.

l = 2 m_l = −2, −1, 0, +1, +2 ; five 4d orbitals.

l = 3 m_l = -3, −2, −1, 0, +1, +2 , +3; seven 4f orbitals

Over all Sixteen orbitals.

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